



SAMPLE CALCULATION

Induction Machine Working As Motor (Set No. __)

$$V_{dc} = \text{_____} V, \quad I_{dc} = \text{_____} A,$$

$$V_{ac} = \text{_____} V, \quad I_{ac} = \text{_____} A, \quad W_{ph} = \text{_____} W$$

$$\text{Time for 10 oscillations} = \text{_____} \text{ sec}$$

$$N_s = \text{_____} \text{ rpm}$$

$$\text{Rotor frequency, } f_2 = 10/T = \text{_____} \text{ Hz}$$

$$\text{Input} = 3 W_{ph} = \text{_____} W$$

$$\text{Output} = V_{dc} \times I_{dc} = \text{_____} W$$

$$\% \text{ Slip} = \frac{f_2}{f} \times 100 = \text{_____} \%$$

$$\text{Speed, } N = (1-s) \times N_s = \text{_____} \text{ rpm}$$

$$\text{Efficiency} = \frac{\text{output}}{\text{input}} \times 100 = \text{_____} \%$$

Induction Machine Working As Generator (Set No. __)

$$V_{dc} = \text{_____} V, \quad I_{dc} = \text{_____} A,$$

$$V_{ac} = \text{_____} V, \quad I_{ac} = \text{_____} A, \quad W_{ph} = \text{_____} W$$

$$\text{Time for 10 oscillations} = \text{_____} \text{ sec}$$

$$N_s = \text{_____} \text{ rpm}$$

$$\text{Rotor frequency, } f_2 = 10/T = \text{_____} \text{ Hz}$$

$$\text{Input} = V_{dc} \times I_{dc} = \text{_____} W$$

$$\text{Output} = 3 W_{ph} = \text{_____} W$$

$$\% \text{ Slip} = -\frac{f_2}{f} \times 100 = \text{_____} \% \text{ (Note : Slip is negative)}$$

$$\text{Speed, } N = (1-s) \times N_s = \text{_____} \text{ rpm (Note : Speed will be above synchronous speed, 1000rpm)}$$

$$\text{Efficiency} = \frac{\text{output}}{\text{input}} \times 100 = \text{_____} \%$$

From 3 W_{ph} Vs % slip characteristics,

$$\text{Hysteresis power} = \frac{AB}{2} = \text{_____} W$$